JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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An Autonomous Institute, with NAAC "A" Grade

Department of CSE

"A place to learn; A chance to grow"

2024-25 (Even Semester)

CLASS TEST 1

Course: B. Tech. in Computer Science & Engineering

Subject: Design & Analysis of Algorithm

Subject Code: CS5T002

Year/Semester: IV Semester (Second Year)

Max Marks: 20 **Date: 03/02/2025** **Duration:- 1 Hr**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO1: Recall basic concepts of algorithm in analysis and Design of algorithms.
CO2: Examine Recurrence relations, solutions of recurrence of searching sorting methods
CO3: Apply Greedy methods used for analysis and Design of Algorithm
CO4: Analyse Dynamic Programming concepts in designing algorithm
CO5: Evaluate advanced techniques and tools available for algorithm analysis and development

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Algorithm.	L1/CO1	1
Q.2	Discuss the term growth rate of function.	L2/CO1	1

Q.3	Find time complexity of given code.
	#include <*stdio.h*>
	void main()
	{
	int i, n = 8;
	for (i = 1; i <= n; i++) {
	printf("Hello World !!!\n");
	}
	}
		L1/CO2	1

Q.4	Explain Time and Space Complexity.	L2/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Write the features or characteristics of Algorithm.	L3/CO1	2
Q.6	Explain the different Asymptotic Notations.	L2/CO2	2
Q.7	Find out the space complexity of following Code. #include <iostream> using namespace std; int sum(int a , int b)	L1/CO2	2